

PC1, PC3, PC5 в VLAN 20

PC2, PC4, PC6 в VLAN 33

VLAN 333 — Native VLAN для транковых портов между коммутаторами

Коммутаторы соединены между собой (скорее всего звездой или цепочкой).

1. Настройка VLAN и транков на всех коммутаторах

```
Switch>enable
Switch#confi
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Layer2Switch-1
Layer2Switch-1(config)#vlan 20
Layer2Switch-1(config-vlan)#name VLAN20
Layer2Switch-1(config-vlan)#vlan 33
Layer2Switch-1(config-vlan)#name VLAN33
Layer2Switch-1(config-vlan)#vlan 333
Layer2Switch-1(config-vlan)#name NATIVE_VLAN
Layer2Switch-1(config-vlan)#interface
Layer2Switch-1(config-vlan)#interface Gi
Layer2Switch-1(config-vlan)#end
Layer2Switch-1#
*Dec 26 08:05:49.027: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-1#
```

Layer2Switch-1 - PuTTY

```
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet0/1
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface G
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet0/2
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet0/3
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet1/0
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet1/1
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inter
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet1/2
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#exit
Layer2Switch-1(config)#inte
Layer2Switch-1(config)#interface Gi
Layer2Switch-1(config)#interface GigabitEthernet1/3
Layer2Switch-1(config-if)#switchport trunk native vlan 333
Layer2Switch-1(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-1(config-if)#no shutdown
Layer2Switch-1(config-if)#end
Layer2Switch-1#write
*Dec 26 08:21:11.607: %SYS-5-CONFIG_I: Configured from console by console
Building configuration...
Compressed configuration from 5855 bytes to 2245 bytes[OK]
Layer2Switch-1#
*Dec 26 08:21:22.733: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:21:23.560: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-1#write memory
Building configuration...
Compressed configuration from 5855 bytes to 2245 bytes[OK]
Layer2Switch-1#
*Dec 26 08:21:29.629: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:21:30.435: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.[]
```

enable

configure terminal

hostname Layer2Switch-1

vlan 20

name VLAN20

```
vlan 33
```

```
name VLAN33
```

```
vlan 333
```

```
name NATIVE_VLAN
```

Транк-порты к другим коммутаторам (пример: к Switch-2 и Switch-3)

```
interface GigabitEthernet 0/1
```

```
switchport mode trunk
```

```
switchport trunk native vlan 333
```

```
switchport trunk allowed vlan 20,33,333
```

```
no shutdown
```

```
interface GigabitEthernet 0/2
```

```
switchport mode trunk
```

```
switchport trunk native vlan 333
```

```
switchport trunk allowed vlan 20,33,333
```

```
no shutdown
```

```
end
```

```
write memory
```

```
Layer2Switch-2
```

```
*****
Switch>enable
Switch#confi
Switch#configure terminal
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname L
Switch(config)#hostname Layer2Switch-2
Layer2Switch-2(config)#vlan 20
Layer2Switch-2(config-vlan)#name VLAN20
Layer2Switch-2(config-vlan)#vlan 33
Layer2Switch-2(config-vlan)#name VLAN33
Layer2Switch-2(config-vlan)#vlan 333
Layer2Switch-2(config-vlan)#name NATIVE_VLAN
Layer2Switch-2(config-vlan)#end
Layer2Switch-2#wrir
*Dec 26 08:27:28.002: %SYS-5-CONFIG_I: Configured from console by console
Translating "wrir"...domain server (255.255.255.255)
(255.255.255.255)
Translating "wrir"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address
Layer2Switch-2#
Layer2Switch-2#
Layer2Switch-2#wwrite memory
^
% Invalid input detected at '^' marker.

Layer2Switch-2#write memory
Building configuration...
Compressed configuration from 5228 bytes to 1996 bytes[OK]
*Dec 26 08:28:15.066: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:28:15.864: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-2#
```

cisco

enable

configure terminal

hostname Layer2Switch-2

```

Layer2Switch-3(config-if)#
*Dec 26 08:43:02.576: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS:
-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->
-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS:
-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32
*Dec 26 08:43:02.577: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vtpvlan_bitlist_test(&pd->ope
-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3
*Dec 26 08:43:02.577: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz
Layer2Switch-3(config-if)#
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 20
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet1/1
Layer2Switch-3(config-if)#switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 33
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/0
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inte
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/1
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/2
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/3
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#end
Layer2Switch-3#wri
*Dec 26 08:46:40.299: %SYS-5-CONFIG_I: Configured from console by cons
Layer2Switch-3#write memory
Building configuration...
Compressed configuration from 5391 bytes to 2095 bytes[OK]
Layer2Switch-3#
*Dec 26 08:46:54.158: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:46:54.910: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-3#

```

vlan 20

name VLAN20

vlan 33

name VLAN33

vlan 333

name NATIVE_VLAN

```
interface GigabitEthernet 1/0
switchport mode access
switchport access vlan 20
no shutdown
```

```
interface GigabitEthernet 1/1
switchport mode access
switchport access vlan 33
no shutdown
```

Транк-порты к другим коммутаторам

```
interface GigabitEthernet 0/0
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,33,333
no shutdown
```

```
interface GigabitEthernet 0/1
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,33,333
no shutdown
```

```
end
```

```
write memory
```

Layer2Switch-3

cisco
enable
configure terminal
hostname Layer2Switch-3

vlan 20
 name VLAN20
vlan 33
 name VLAN33
vlan 333
 name NATIVE_VLAN

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Layer2Switch-3
Layer2Switch-3(config)#vlan 20
Layer2Switch-3(config-vlan)#name VLAN20
Layer2Switch-3(config-vlan)#vlan 33
Layer2Switch-3(config-vlan)#name VLAN33
Layer2Switch-3(config-vlan)#vlan 333
Layer2Switch-3(config-vlan)#name NATIVE_VLAN
Layer2Switch-3(config-vlan)#end
Layer2Switch-3#
*Dec 26 08:51:49.047: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-3#write memory
Building configuration...
Compressed configuration from 5037 bytes to 1931 bytes[OK]
Layer2Switch-3#
*Dec 26 08:52:04.419: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:52:05.175: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-3#
```

```

Layer2Switch-3 - PuTTY
Layer2Switch-3(config-if)#
*Dec 26 08:43:02.576: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../
-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->glob
-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../
-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A
*Dec 26 08:43:02.577: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 3291
*Dec 26 08:43:02.577: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vtpvlan_bitlist_test(&pd->operInf
-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D
*Dec 26 08:43:02.577: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 3291
Layer2Switch-3(config-if)#
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 20
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet1/1
Layer2Switch-3(config-if)#switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 33
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/0
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inte
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/1
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/2
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/3
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#end
Layer2Switch-3#wri
*Dec 26 08:46:40.299: %SYS-5-CONFIG_I: Configured from console by cons
Layer2Switch-3#write memory
Building configuration...
Compressed configuration from 5391 bytes to 2095 bytes[OK]
Layer2Switch-3#
*Dec 26 08:46:54.158: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:46:54.910: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-3#

```

interface GigabitEthernet 1/0

switchport mode access

switchport access vlan 20

no shutdown

interface GigabitEthernet 1/1

switchport mode access

switchport access vlan 33

no shutdown

Транки к соседним коммутаторам

interface GigabitEthernet 0/0

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,33,333

no shutdown

interface GigabitEthernet0/1

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,33,333

no shutdown

end

write memory

Layer2Switch-4

```
Switch>enable
Switch#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#hostname Layer2Switch-3
Layer2Switch-3(config)#vlan 20
Layer2Switch-3(config-vlan)#name VLAN20
Layer2Switch-3(config-vlan)#vlan 33
Layer2Switch-3(config-vlan)#name VLAN33
Layer2Switch-3(config-vlan)#vlan 333
Layer2Switch-3(config-vlan)#name NATIVE_VLAN
Layer2Switch-3(config-vlan)#end
Layer2Switch-3#
*Dec 26 08:51:49.047: %SYS-5-CONFIG_I: Configured from console by cons
Layer2Switch-3#write memory
Building configuration...
Compressed configuration from 5037 bytes to 1931 bytes[OK]
Layer2Switch-3#
*Dec 26 08:52:04.419: %GRUB-5-CONFIG_WRITING: GRUB configuration is be
*Dec 26 08:52:05.175: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was w
```

```

Layer2Switch-4 - PuTTY
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet1/0
Layer2Switch-3(config-if)#switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 20
Layer2Switch-3(config-if)#
*Dec 26 08:53:02.490: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/
-Traceback= 1C492C9z F385ABz FA54E0z FA5617z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290
*Dec 26 08:53:02.491: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vlan_test_portlist(vlan, pd->globalNumbe
-Traceback= 1C492C9z F385ABz FA578Bz F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 329
*Dec 26 08:53:02.491: %PM-3-INTERNALERROR: Port Manager Internal Software Error (vlan > 0 && vlan < PM_MAX_VLANS: ../switch/
-Traceback= 1C492C9z F385ABz FA58F1z FA5665z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290
*Dec 26 08:53:02.491: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196BF98z FA5699z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 32
*Dec 26 08:53:02.491: %PM-3-INTERNALERROR: Port Manager Internal Software Error (pm_vtpvlan_bitlist_test(&pd->operInfo.trunk
-Traceback= 1C492C9z F385ABz FA5700z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 3290D9Bz 329
*Dec 26 08:53:02.491: %BIT-4-OUTOFRANGE: bit 0 is not in the expected range of 1 to 4095
-Traceback= 1C492C9z 196C71Fz 196C008z FA56A7z F9398Bz F93DD7z 3013F8Fz F9CD75z F42DD8z 10C9EAEz F48DBDz F5DE6Cz 32915A5z 32
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet1/1
Layer2Switch-3(config-if)# switchport mode access
Layer2Switch-3(config-if)#switchport access vlan 33
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/0
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/1
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/2
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#exit
Layer2Switch-3(config)#inter
Layer2Switch-3(config)#interface Gi
Layer2Switch-3(config)#interface GigabitEthernet0/3
Layer2Switch-3(config-if)#switchport trunk native vlan 333
Layer2Switch-3(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-3(config-if)#no shutdown
Layer2Switch-3(config-if)#end
Layer2Switch-3#
*Dec 26 08:56:57.347: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-3#write memory
Building configuration...
Compressed configuration from 5391 bytes to 2096 bytes[OK]
Layer2Switch-3#
*Dec 26 08:57:18.049: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 08:57:18.803: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-3#

```

cisco

enable

configure terminal

hostname Layer2Switch-4

vlan 20

name VLAN20

vlan 33

```
name VLAN33
vlan 333
name NATIVE_VLAN
```

```
interface GigabitEthernet 1/0
switchport mode access
switchport access vlan 20
no shutdown
```

```
interface GigabitEthernet 1/1
switchport mode access
switchport access vlan 33
no shutdown
```

```
Транки
interface GigabitEthernet 0/0
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,33,333
no shutdown
```

```
interface GigabitEthernet 0/1
switchport mode trunk
switchport trunk native vlan 333
switchport trunk allowed vlan 20,33,333
no shutdown
```

```
end
write memory
```

Layer2Switch-5

cisco

enable

configure terminal

hostname Layer2Switch-5

```
Switch>enable
Switch#confi
Switch#configure ter
Switch#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#hostname Layer2Switch-5
Layer2Switch-5(config)#vlan 20
Layer2Switch-5(config-vlan)#name VLAN20
Layer2Switch-5(config-vlan)#vlan 33
Layer2Switch-5(config-vlan)#
Layer2Switch-5(config-vlan)#name VLAN33
Layer2Switch-5(config-vlan)#vlan 333
Layer2Switch-5(config-vlan)#name NATIVE_VLAN
Layer2Switch-5(config-vlan)#end
Layer2Switch-5#w
*Dec 26 09:00:22.691: %SYS-5-CONFIG_I: Configured from console by consol
Layer2Switch-5#write memory
Building configuration...
Compressed configuration from 5030 bytes to 1928 bytes[OK]
Layer2Switch-5#
*Dec 26 09:00:35.711: %GRUB-5-CONFIG_WRITING: GRUB configuration is being updated on disk. Please wait...
*Dec 26 09:00:36.459: %GRUB-5-CONFIG_WRITTEN: GRUB configuration was written to disk successfully.
Layer2Switch-5#
```

Port	Name	Status	Vlan	Duplex	Speed	Type
Gi0/0		connected	trunk	auto	auto	unknown
Gi0/1		connected	trunk	auto	auto	unknown
Gi0/2		connected	trunk	auto	auto	unknown
Gi0/3		connected	trunk	auto	auto	unknown
Gil/0		connected	1	auto	auto	unknown
Gil/1		connected	1	auto	auto	unknown

```

Layer2Switch-5#confi
Layer2Switch-5#configure ter
Layer2Switch-5#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
Layer2Switch-5(config)#inter
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet1/0
Layer2Switch-5(config-if)#switchport mode access
Layer2Switch-5(config-if)#switchport access vlan 20
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#inter
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet1/1
Layer2Switch-5(config-if)#switchport mode access\
^
% Invalid input detected at '^' marker.

Layer2Switch-5(config-if)#switchport mode access
Layer2Switch-5(config-if)#switchport access vlan 33
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#inter
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet0/0
Layer2Switch-5(config-if)#switchport trunk native vlan 333
Layer2Switch-5(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#inter
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet0/1
Layer2Switch-5(config-if)#switchport trunk native vlan 333
Layer2Switch-5(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#int
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet0/2
Layer2Switch-5(config-if)#switchport trunk native vlan 333
Layer2Switch-5(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#exit
Layer2Switch-5(config)#inter
Layer2Switch-5(config)#interface Gi
Layer2Switch-5(config)#interface GigabitEthernet0/2
Layer2Switch-5(config-if)#switchport trunk native vlan 333
Layer2Switch-5(config-if)#switchport trunk allowed vlan 20,33,333
Layer2Switch-5(config-if)#no shutdown
Layer2Switch-5(config-if)#end
Layer2Switch-5#
*Dec 26 09:04:53.310: %SYS-5-CONFIG_I: Configured from console by console
Layer2Switch-5#write memory
Building configuration...

```

vlan 20

name VLAN20

vlan 33

name VLAN33

vlan 333

name NATIVE_VLAN

interface GigabitEthernet 1/0

switchport mode access

switchport access vlan 20

no shutdown

interface GigabitEthernet 1/1

switchport mode access

switchport access vlan 33

no shutdown

Транки

interface GigabitEthernet 0/0

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,33,333

no shutdown

interface GigabitEthernet 0/1

switchport mode trunk

switchport trunk native vlan 333

switchport trunk allowed vlan 20,33,333

no shutdown

end

write memory

еще на каждом интерфейсе где установлен mode trunk была выполнена команда

switchport trunk encapsulation dot1q

2. Проверка доступности внутри VLAN и между VLAN

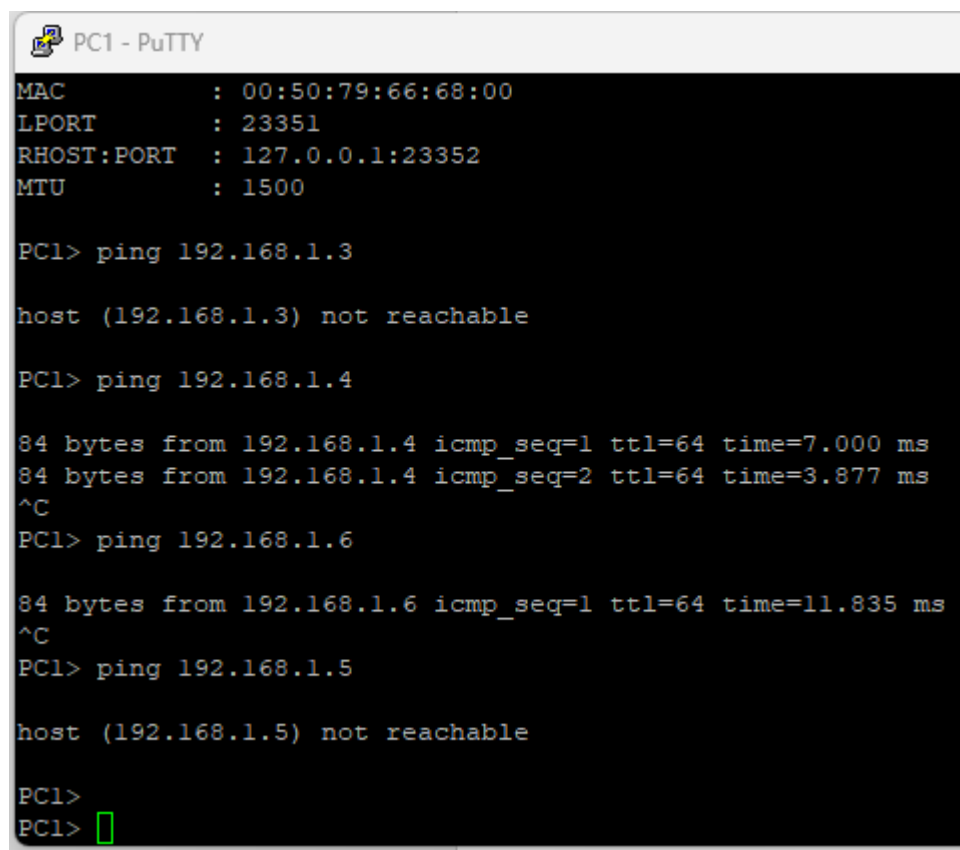
Команды проверки на коммутаторах:

show vlan brief

show interfaces trunk

show running-config

Проверка с ПК:



```
PC1 - PuTTY
MAC       : 00:50:79:66:68:00
LPORT     : 23351
RHOST:PORT : 127.0.0.1:23352
MTU       : 1500

PC1> ping 192.168.1.3

host (192.168.1.3) not reachable

PC1> ping 192.168.1.4

84 bytes from 192.168.1.4 icmp_seq=1 ttl=64 time=7.000 ms
84 bytes from 192.168.1.4 icmp_seq=2 ttl=64 time=3.877 ms
^C
PC1> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=11.835 ms
^C
PC1> ping 192.168.1.5

host (192.168.1.5) not reachable

PC1>
PC1> █
```

```
PC2> ping 192.168.1.1
```

```
host (192.168.1.1) not reachable
```

```
PC2> ping 192.168.1.3
```

```
192.168.1.3 icmp_seq=1 ttl=64 time=0.001 ms  
192.168.1.3 icmp_seq=2 ttl=64 time=0.001 ms  
192.168.1.3 icmp_seq=3 ttl=64 time=0.001 ms  
192.168.1.3 icmp_seq=4 ttl=64 time=0.001 ms  
192.168.1.3 icmp_seq=5 ttl=64 time=0.001 ms
```

```
PC2> ping 192.168.1.2
```

```
host (192.168.1.2) not reachable
```

```
PC2> ping 192.168.1.4
```

```
host (192.168.1.4) not reachable
```

```
PC2> ping 192.168.1.5
```

```
84 bytes from 192.168.1.5 icmp_seq=1 ttl=64 time=7.097 ms  
84 bytes from 192.168.1.5 icmp_seq=2 ttl=64 time=20.503 ms  
84 bytes from 192.168.1.5 icmp_seq=3 ttl=64 time=11.005 ms  
84 bytes from 192.168.1.5 icmp_seq=4 ttl=64 time=12.089 ms  
84 bytes from 192.168.1.5 icmp_seq=5 ttl=64 time=9.501 ms
```

```
PC2> ping 192.168.1.6
```

```
host (192.168.1.6) not reachable
```

```
PC2> 
```



```
PC3 - PuTTY

PC3> ping 192.168.1.2

84 bytes from 192.168.1.2 icmp_seq=1 ttl=64 time=10.314 ms
84 bytes from 192.168.1.2 icmp_seq=2 ttl=64 time=5.819 ms
^C
PC3> ping 192.168.1.3

host (192.168.1.3) not reachable

PC3> ping 192.168.1.5

host (192.168.1.5) not reachable

PC3> ping 192.168.1.6

84 bytes from 192.168.1.6 icmp_seq=1 ttl=64 time=11.155 ms
84 bytes from 192.168.1.6 icmp_seq=2 ttl=64 time=15.084 ms
^C
PC3> 
```

```
PC4 - PuTTY

PC4> ping 192.168.1.2

host (192.168.1.2) not reachable

PC4> ping 192.168.1.3

84 bytes from 192.168.1.3 icmp_seq=1 ttl=64 time=11.356 ms
84 bytes from 192.168.1.3 icmp_seq=2 ttl=64 time=7.626 ms
^C
PC4> ping 192.168.1.4

host (192.168.1.4) not reachable

PC4> ping 192.168.1.6

host (192.168.1.6) not reachable

PC4> ping 192.168.1.7

84 bytes from 192.168.1.7 icmp_seq=1 ttl=64 time=14.832 ms
84 bytes from 192.168.1.7 icmp_seq=2 ttl=64 time=10.615 ms
84 bytes from 192.168.1.7 icmp_seq=3 ttl=64 time=12.829 ms
84 bytes from 192.168.1.7 icmp_seq=4 ttl=64 time=11.793 ms
^C
PC4> 
```

ping <IP_в_той_же_VLAN> → должен быть успех

ping <IP_в_другой_VLAN> → должен быть неуспех (без маршрутизатора)

3. Перехват пакетов в Wireshark

Id = 20

Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Man

No.	Time	Source	Destination	Protocol	Length	Info
34	50.441680	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
35	52.563656	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
36	53.580112	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	60	Dynamic Trunk Protocol
37	53.580153	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	90	Dynamic Trunk Protocol
38	54.673013	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
39	55.673537	Private_66:68:00	Broadcast	ARP	64	Who has 192.168.1.4? Tell 192.168.1.2
40	55.679004	Private_66:68:02	Private_66:68:00	ARP	64	192.168.1.4 is at 00:50:79:66:68:02
41	55.680061	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xec9b, seq=1/256, ttl=64 (reply in 42)
42	55.691059	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xec9b, seq=1/256, ttl=64 (request in 41)
43	56.691988	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xed9b, seq=2/512, ttl=64 (reply in 44)
44	56.696642	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xed9b, seq=2/512, ttl=64 (request in 43)
45	56.778736	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
46	57.697307	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xee9b, seq=3/768, ttl=64 (reply in 47)
47	57.705627	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xee9b, seq=3/768, ttl=64 (request in 46)
48	58.447159	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	CDP	432	Device ID: Layer2Switch-3 Port ID: GigabitEthernet1/0
49	58.705936	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xef9b, seq=4/1024, ttl=64 (reply in 50)
50	58.713308	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xef9b, seq=4/1024, ttl=64 (request in 49)
51	58.901381	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
52	59.714458	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xf09b, seq=5/1280, ttl=64 (reply in 53)
53	59.721802	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xf09b, seq=5/1280, ttl=64 (request in 52)
54	60.464803	0c:9f:57:66:00:04	0c:9f:57:66:00:04	LOOP	60	Reply
55	61.007167	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
56	63.099462	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
57	65.227824	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
58	67.336430	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
59	69.466615	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
60	70.553545	0c:9f:57:66:00:04	0c:9f:57:66:00:04	LOOP	60	Reply
61	71.569868	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
62	73.678833	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005

Frame 52: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:00 (00:50:79:66:68:00), Dst: Private_66:68:02 (00:50:79:66:68:02)

Internet Protocol Version 4, Src: 192.168.1.2, Dst: 192.168.1.4

Internet Control Message Protocol

0000 00 50 79 66 68 02 00 50 79 66 68 00 00 00 45 00 Pyfh P yfh...E
0010 00 54 9b f0 00 00 40 01 5b 62 c0 a8 01 02 c0 a8 T...@ [b.....
0020 01 04 08 00 2f 6b f0 9b 00 05 08 09 0a 0b 0c 0d .../k.....
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ..!#\$%&'()*+,-.
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!#\$%&'()*+,-./012345 6789:;<=
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
0060 3e 3f >?

Захват с Standard input [Layer2Switch-3 Ethernet4 to PC1 Ethernet0]

Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Man

No.	Time	Source	Destination	Protocol	Length	Info
43	56.691988	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xed9b, seq=2/512, ttl=64 (reply in 44)
44	56.696642	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xed9b, seq=2/512, ttl=64 (request in 43)
45	56.778736	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
46	57.697307	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xee9b, seq=3/768, ttl=64 (reply in 47)
47	57.705627	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xee9b, seq=3/768, ttl=64 (request in 46)
48	58.447159	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	CDP	432	Device ID: Layer2Switch-3 Port ID: GigabitEthernet1/0
49	58.705936	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xef9b, seq=4/1024, ttl=64 (reply in 50)
50	58.713308	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xef9b, seq=4/1024, ttl=64 (request in 49)
51	58.901381	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
52	59.714458	192.168.1.2	192.168.1.4	ICMP	98	Echo (ping) request id=0xf09b, seq=5/1280, ttl=64 (reply in 53)
53	59.721802	192.168.1.4	192.168.1.2	ICMP	98	Echo (ping) reply id=0xf09b, seq=5/1280, ttl=64 (request in 52)
54	60.464803	0c:9f:57:66:00:04	0c:9f:57:66:00:04	LOOP	60	Reply
55	61.007167	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
56	63.099462	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
57	65.227824	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
58	67.336430	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
59	69.466615	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
60	70.553545	0c:9f:57:66:00:04	0c:9f:57:66:00:04	LOOP	60	Reply
61	71.569868	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
62	73.678833	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
63	75.780801	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
64	77.856835	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
65	79.951379	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
66	80.617747	0c:9f:57:66:00:04	0c:9f:57:66:00:04	LOOP	60	Reply
67	82.054997	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
68	83.824530	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	60	Dynamic Trunk Protocol
69	83.824606	0c:9f:57:66:00:04	CDP/VTP/DTP/PAgP/UD...	DTP	90	Dynamic Trunk Protocol
70	84.153758	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005
71	86.288792	0c:9f:57:66:00:04	Nearest-Customer-Br...	STP	60	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 4 Port = 0x8005

Frame 53: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0

Ethernet II, Src: Private_66:68:02 (00:50:79:66:68:02), Dst: Private_66:68:00 (00:50:79:66:68:00)

Internet Protocol Version 4, Src: 192.168.1.4, Dst: 192.168.1.2

Internet Control Message Protocol

0000 00 50 79 66 68 00 00 50 79 66 68 02 00 00 45 00 Pyfh P yfh...E
0010 00 54 9b f0 00 00 40 01 5b 62 c0 a8 01 04 c0 a8 T...@ [b.....
0020 01 02 00 00 37 6b f0 9b 00 05 08 09 0a 0b 0c 0d ...7k.....
0030 0e 0f 10 11 12 13 14 15 16 17 18 19 1a 1b 1c 1d ..!#\$%&'()*+,-.
0040 1e 1f 20 21 22 23 24 25 26 27 28 29 2a 2b 2c 2d ...!#\$%&'()*+,-./012345 6789:;<=
0050 2e 2f 30 31 32 33 34 35 36 37 38 39 3a 3b 3c 3d
0060 3e 3f >?

```
Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка
vlan.id = 20

No. Time Source Destination Protocol Length Info
---
146 75.613020 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
151 77.717113 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
155 79.812666 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
158 81.950661 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
162 84.078740 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
165 86.179079 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
169 88.269148 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
173 90.379018 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
176 92.496900 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
179 94.603153 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
183 96.719160 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
186 98.842055 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
-- 191 100.916272 192.168.1.2 192.168.1.4 ICMP 102 Echo (ping) request id=0x1d9c, seq=1/256, ttl=64 (reply in 192)
-- 192 100.919662 192.168.1.4 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1d9c, seq=1/256, ttl=64 (request in 191)
193 100.939541 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
196 101.920610 192.168.1.2 192.168.1.4 ICMP 102 Echo (ping) request id=0x1e9c, seq=2/512, ttl=64 (reply in 197)
197 101.926805 192.168.1.4 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1e9c, seq=2/512, ttl=64 (request in 196)
198 102.929163 192.168.1.2 192.168.1.4 ICMP 102 Echo (ping) request id=0x1f9c, seq=3/768, ttl=64 (reply in 199)
199 102.932458 192.168.1.4 192.168.1.2 ICMP 102 Echo (ping) reply id=0x1f9c, seq=3/768, ttl=64 (request in 198)
200 103.060895 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
203 103.934622 192.168.1.2 192.168.1.4 ICMP 102 Echo (ping) request id=0x209c, seq=4/1024, ttl=64 (reply in 204)
204 103.939391 192.168.1.4 192.168.1.2 ICMP 102 Echo (ping) reply id=0x209c, seq=4/1024, ttl=64 (request in 203)
205 104.941040 192.168.1.2 192.168.1.4 ICMP 102 Echo (ping) request id=0x219c, seq=5/1280, ttl=64 (reply in 206)
206 104.949181 192.168.1.4 192.168.1.2 ICMP 102 Echo (ping) reply id=0x219c, seq=5/1280, ttl=64 (request in 205)
207 105.131619 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
211 107.219662 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
215 109.331547 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
219 111.446241 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
223 113.537225 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
226 115.627914 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
230 117.728885 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
234 119.808957 0c:2c:33:d0:00:02 PVST+ STP 68 Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003

> Frame 192: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface -, id 0
> Eth II, Src: Private GigabitEthernet0/20 (00:50:79:66:68:02), Dst: Private GigabitEthernet0/20 (00:50:79:66:68:00)
> 802.1 Virtual LAN, Prio: 0, DEI: 0, ID: 20
> Internet Protocol Version 4, Src: 192.168.1.4, Dst: 192.168.1.2
> Internet Control Message Protocol

0000 00 50 79 66 68 00 00 50 79 66 68 02 81 00 00 14 -Pyfh-P yfh
0010 00 00 00 00 54 0c 1d 00 00 40 01 50 35 c0 a0 -E.T...Q[5
0020 01 04 c0 a8 01 02 00 00 00 00 00 00 00 00 00 00 -
0030 0a 00 0c 0d 0e 0f 10 11 12 13 14 15 16 17 18 19 -
0040 1a 1b 1c 1d 1e 1f 20 21 22 23 24 25 26 27 28 29 -.....! "#$%&'()
0050 2a 2b 2c 2d 2e 2f 30 31 32 33 34 35 36 37 38 39 -*,+,-./01 23456789
0060 3a 3b 3c 3d 3e 3f ;:<=>?
```

С теґом

Захват с Standard input [Layer2Switch-1 Ethernet2 to Layer2Switch-3 Ethernet0]
Файл Правка Вид Запуск Захват Анализ Статистика Телефония Беспроводная связь Инструменты Справка

Вид: hex ==20
100.0.0.0/8

No.	Time	Source	Destination	Protocol	Length	Info
146	75.613020	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
151	77.717113	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
155	79.812666	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
158	81.950661	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
162	84.078740	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
165	86.179079	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
168	88.269148	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
173	90.379018	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
176	92.496906	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
179	94.603153	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
183	96.719168	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
186	98.842955	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
---	191.100.916272	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x1d9c, seq=1/256, ttl=64 (reply in 192)
---	192.100.919062	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) reply id=0x1d9c, seq=1/256, ttl=64 (request in 191)
193	100.939541	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
196	101.920610	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x1e9c, seq=2/512, ttl=64 (reply in 197)
197	101.926805	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) reply id=0x1e9c, seq=2/512, ttl=64 (request in 196)
198	102.929163	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0xf19c, seq=3/768, ttl=64 (reply in 199)
199	102.932458	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) reply id=0xf19c, seq=3/768, ttl=64 (request in 198)
200	103.060855	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
203	103.934622	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x209c, seq=4/1024, ttl=64 (reply in 204)
204	103.939301	192.168.1.2	192.168.1.2	ICMP	102	Echo (ping) reply id=0x209c, seq=4/1024, ttl=64 (request in 203)
205	104.941040	192.168.1.2	192.168.1.4	ICMP	102	Echo (ping) request id=0x219c, seq=5/1280, ttl=64 (reply in 206)
206	104.949101	192.168.1.2	192.168.1.2	ICMP	102	Echo (ping) reply id=0x219c, seq=5/1280, ttl=64 (request in 205)
207	105.131619	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
211	107.919662	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
215	109.331547	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
219	111.446241	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
223	113.537225	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
226	115.627914	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
230	117.720085	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003
234	119.008967	0c:2c:33:d0:00:02	PVST+	STP	68	Conf. Root = 32768/20/0c:2c:33:d0:00:00 Cost = 0 Port = 0x0003

▶ Frame 191: Packet, 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface - , id 0

▶ Ethernet II, Src: Private, 66:68:00: (00:50:79:66:

Id == 333

The screenshot displays the Wireshark network protocol analyzer interface. At the top, the title bar reads "Захват с Standard input (Layer2Switch-3 Ethernet0 to PC2 Ethernet0)". The main menu includes options like Файл, Правка, Вид, Запуск, Захват, Анализ, Статистика, Телефония, Беспроводная связь, Инструменты, and Справка.

The interface is divided into three main sections:

- Packets List Panel (Left):** Shows a list of captured packets. Packet 18 is highlighted, which is an ICMP Echo (ping) request from 192.168.1.3 to 192.168.1.5.
- Packet Details Panel (Middle):** Provides a hierarchical view of the selected packet's structure. For packet 18, it shows Ethernet II, Internet Protocol Version 4, and ICMP Echo (ping) request details.
- Packet Bytes Panel (Right):** Displays the raw hexadecimal and ASCII data of the selected packet. It shows the start of an ICMP Echo (ping) request with hex values like 00 50 79 66 68 03 00 50.

The bottom status bar indicates: "Frame 13: Packet, 98 bytes on wire (784 bits), 98 bytes captured (784 bits) on interface -, id 0".

На транковом порту: будут видны теги 802.1Q (VLAN 20, 33), кроме пакетов Native VLAN 333 (без тегов).

4. Сохранение конфигураций

На каждом коммутаторе:

